Basic matrix operations such as addition, multiplication, inverse, transpose,

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determinant, submatrix, compatibility, element-wise multiplication etc.

```
% Author: Shushank
% Date: 2026-08-18

close all; clc;
```

Matrices

```
A = [1 2; 3 4];
B = [5 6; 7 8];
```

Basic Matrix Operations

```
addition_result       = A + B;
subtraction_result    = A - B;
multiplication_result = A * B;
elementwise_result    = A .* B;

inverse_A = inv(A);
inverse_B = inv(B);

transpose_A = A';
transpose_B = B';

determinant_A = det(A);
determinant_B = det(B);
```

```
rank_A = rank(A);
rank_B = rank(B);

submatrix_A = A(1:2,1);
[mA,nA] = size(A);
[mB,nB] = size(B);
is_compatible = (nA == mB);
```

Eigenvalues and Eigenvectors

```
[eigenvectors_A,eigenvalues_A] = eig(A);
[eigenvectors_B,eigenvalues_B] = eig(B);

eigenvalues_A = diag(eigenvalues_A);
eigenvalues_B = diag(eigenvalues_B);
```

Characteristic Polynomial and Equation

```
syms lambda

characteristic_polynomial_A = poly(A);
characteristic_polynomial_B = poly(B);

characteristic_equation_A = ...
    det(lambda*eye(size(A))-A) == 0;
characteristic_equation_B = ...
    det(lambda*eye(size(B))-B) == 0;
```

Cayley-Hamilton Theorem

```
cA = characteristic_polynomial_A;
cB = characteristic_polynomial_B;

cayley_hamilton_A = cA(1)*A^2 + cA(2)*A + cA(3)*eye(size(A));
cayley_hamilton_B = cB(1)*B^2 + cB(2)*B + cB(3)*eye(size(B));

verified_A = isequal(round(cayley_hamilton_A,10),zeros(size(A)));
verified_B = isequal(round(cayley_hamilton_B,10),zeros(size(B)));
```

System of Linear Equations

$x + 2y = 5$ $3x + 4y = 11$

```
C = [5;11];
solution_X = A\C;
x = solution_X(1);
y = solution_X(2);
verification = A*solution_X;
```

Display Results

```
fprintf('\n========== MATRIX OPERATIONS ==========\n');

disp('A ='); disp(A);
disp('B ='); disp(B);

disp('A + B ='); disp(addition_result);
disp('A - B ='); disp(subtraction_result);
disp('A * B ='); disp(multiplication_result);
disp('A .* B ='); disp(elementwise_result);

disp('Inverse of A ='); disp(inverse_A);
disp('Inverse of B ='); disp(inverse_B);

disp('Transpose of A ='); disp(transpose_A);
disp('Transpose of B ='); disp(transpose_B);

fprintf('det(A) = %.2f\n',determinant_A);
fprintf('det(B) = %.2f\n',determinant_B);
fprintf('rank(A) = %d\n',rank_A);
fprintf('rank(B) = %d\n',rank_B);

disp('Submatrix of A ='); disp(submatrix_A);
fprintf('Size of A = %d x %d\n',mA,nA);
fprintf('Size of B = %d x %d\n',mB,nB);
fprintf('A and B compatible for multiplication: %d\n',is_compatible);

fprintf('\n========== EIGENVALUES / EIGENVECTORS ==========\n');

disp('Eigenvalues of A ='); disp(eigenvalues_A);
disp('Eigenvectors of A ='); disp(eigenvectors_A);
disp('Eigenvalues of B ='); disp(eigenvalues_B);
disp('Eigenvectors of B ='); disp(eigenvectors_B);

fprintf('\n========== CHARACTERISTIC POLYNOMIAL ==========\n');

disp('Coefficients for A ='); disp(characteristic_polynomial_A);
disp('Coefficients for B ='); disp(characteristic_polynomial_B);
disp('Characteristic equation of A ='); disp(characteristic_equation_A);
disp('Characteristic equation of B ='); disp(characteristic_equation_B);

fprintf('\n========== CAYLEY-HAMILTON THEOREM ==========\n');

disp('p(A) ='); disp(cayley_hamilton_A);
fprintf('Theorem verified for A: %d\n',verified_A);

disp('p(B) ='); disp(cayley_hamilton_B);
fprintf('Theorem verified for B: %d\n',verified_B);

fprintf('\n========== LINEAR EQUATIONS ==========\n');

disp('Coefficient matrix A ='); disp(A);
```

```
disp('Right-hand side C ='); disp(C);
disp('Solution [x;y] ='); disp(solution_X);
fprintf('x = %.2f, y = %.2f\n',x,y);
disp('Verification A*X ='); disp(verification);


========== MATRIX OPERATIONS ==========
A =
     1     2
     3     4

B =
     5     6
     7     8

A + B =
     6     8
    10    12

A - B =
    -4    -4
    -4    -4

A * B =
    19    22
    43    50

A .* B =
     5    12
    21    32

Inverse of A =
   -2.0000    1.0000
    1.5000   -0.5000

Inverse of B =
   -4.0000    3.0000
    3.5000   -2.5000

Transpose of A =
     1     3
     2     4

Transpose of B =
     5     7
     6     8

det(A) = -2.00
det(B) = -2.00
rank(A) = 2
rank(B) = 2
Submatrix of A =
     1
     3
```

```
Size of A = 2 x 2
Size of B = 2 x 2
A and B compatible for multiplication: 1

========== EIGENVALUES / EIGENVECTORS ==========
Eigenvalues of A =
   -0.3723
    5.3723

Eigenvectors of A =
   -0.8246   -0.4160
    0.5658   -0.9094

Eigenvalues of B =
   -0.1521
   13.1521

Eigenvectors of B =
   -0.7587   -0.5928
    0.6515   -0.8054


========== CHARACTERISTIC POLYNOMIAL ==========
Coefficients for A =
    1.0000   -5.0000   -2.0000

Coefficients for B =
    1.0000  -13.0000   -2.0000

Characteristic equation of A =
lambda^2 - 5*lambda - 2 == 0

Characteristic equation of B =
lambda^2 - 13*lambda - 2 == 0


========== CAYLEY-HAMILTON THEOREM ==========
p(A) =
   1.0e-15 *

    0.2220         0
         0    0.2220

Theorem verified for A: 1
p(B) =
   1.0e-14 *

    0.2442         0
         0    0.2442

Theorem verified for B: 1

========== LINEAR EQUATIONS ==========
```

```
Coefficient matrix A =
     1     2
     3     4

Right-hand side C =
     5
    11

Solution [x;y] =
     1
     2

x = 1.00, y = 2.00
Verification A*X =
     5
    11
```

Plotting

```
figure;
subplot(1,2,1);
plot(A,'LineWidth',2);
title('Matrix A'); xlabel('Column'); ylabel('Value'); grid on;

subplot(1,2,2);
plot(B,'LineWidth',2);
title('Matrix B'); xlabel('Column'); ylabel('Value'); grid on;

fprintf('\n========== PROGRAM COMPLETED ==========\n');
```

```
========== PROGRAM COMPLETED ==========
```

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